

## ADELBODEN, Switzerland

WMO: 067350

Lat: 46.4917N

Lon: 7.5607E

Elev: 4357

StdP: 12.52

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 8.1        | 11.9 | -1.0                          | 6.1 | 13.6 | 3.2 | 7.6 | 17.6 | 24.1                  | 41.8 | 20.6 | 40.7 | 2.1                   | 190  | 0.583 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.8                   | 76.9            | 61.3 | 73.9 | 60.4 | 70.8 | 59.3 | 63.2                | 73.3 | 61.7 | 71.1 | 60.2 | 68.8 | 7.8                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 59.4                            | 89.0 | 67.7 | 58.0 | 84.5 | 66.4 | 56.6 | 80.3 | 64.9 | 31.0          | 74.0 | 29.8 | 71.2 | 28.7 | 69.2 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 16.5              | 13.3 | 11.3 | DB                         | 3.9 | 81.8               | 4.3 | 2.6                                                | 0.8  | 83.7       | -1.7 | 85.2       | -4.1 | 86.7       | -7.2 | 88.5 |
|                   |      |      | WB                         | 2.8 | 66.7               | 4.1 | 1.5                                                | -0.2 | 67.8       | -2.6 | 68.7       | -4.9 | 69.6       | -7.9 | 70.7 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.2   | 29.7  | 29.6  | 35.8  | 42.9  | 48.6  | 56.5  | 59.2  | 58.3  | 52.1  | 47.0  | 37.6  | 31.9  | (6)  |
|                                                                        | DBStd       | 13.10  | 8.52  | 9.42  | 8.25  | 7.51  | 7.31  | 7.15  | 6.11  | 6.40  | 6.42  | 7.58  | 8.54  | 8.66  | (7)  |
|                                                                        | HDD50       | 3152   | 628   | 570   | 441   | 232   | 115   | 21    | 6     | 8     | 53    | 142   | 374   | 562   | (8)  |
|                                                                        | HDD65       | 7644   | 1093  | 990   | 906   | 664   | 508   | 268   | 197   | 224   | 387   | 559   | 821   | 1027  | (9)  |
|                                                                        | CDD50       | 1030   | 0     | 0     | 1     | 18    | 72    | 215   | 290   | 265   | 116   | 49    | 3     | 1     | (10) |
|                                                                        | CDD65       | 43     | 0     | 0     | 0     | 0     | 0     | 12    | 16    | 15    | 0     | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 244    | 0     | 0     | 0     | 0     | 2     | 76    | 90    | 75    | 1     | 0     | 0     | 0     | (12) |
|                                                                        | CDH80       | 20     | 0     | 0     | 0     | 0     | 0     | 8     | 8     | 4     | 0     | 0     | 0     | 0     | (13) |
|                                                                        | WSAvg       | 3.8    | 3.7   | 3.8   | 4.3   | 4.1   | 4.2   | 4.1   | 3.9   | 3.6   | 3.4   | 3.2   | 3.4   | 3.7   | (14) |
|                                                                        | PrecAvg     | 39.1   | 2.8   | 2.8   | 2.6   | 2.6   | 3.8   | 3.9   | 4.3   | 4.1   | 2.8   | 2.8   | 3.2   | 3.5   | (15) |
| (16) Precipitation                                                     | PrecMax     | 50.1   | 8.3   | 9.8   | 6.4   | 5.3   | 7.2   | 7.4   | 8.0   | 7.7   | 5.6   | 6.1   | 8.3   | 9.1   | (16) |
|                                                                        | PrecMin     | 29.2   | 0.3   | 0.5   | 0.8   | 0.7   | 1.4   | 2.0   | 1.8   | 1.3   | 1.3   | 0.1   | 0.5   | 0.1   | (17) |
|                                                                        | PrecStd     | 5.2    | 1.9   | 2.2   | 1.5   | 1.1   | 1.8   | 1.3   | 1.4   | 1.5   | 1.1   | 1.4   | 1.9   | 2.1   | (18) |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |      |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 51.8  | 53.1  | 57.6  | 66.3  | 72.7  | 80.7  | 80.5  | 80.4  | 71.6  | 67.8  | 58.6  | 54.4  | (19) |
|                                                                        |             | MCWB   | 38.7  | 39.0  | 43.3  | 49.2  | 56.1  | 61.3  | 62.5  | 61.8  | 58.2  | 50.4  | 44.5  | 40.5  | (20) |
|                                                                        | 2%          | DB     | 47.4  | 48.2  | 54.0  | 62.3  | 68.0  | 75.9  | 76.4  | 75.9  | 68.3  | 63.9  | 55.5  | 49.3  | (21) |
|                                                                        |             | MCWB   | 37.5  | 37.1  | 41.5  | 48.1  | 54.1  | 60.9  | 61.0  | 61.5  | 57.5  | 51.1  | 43.6  | 38.0  | (22) |
|                                                                        | 5%          | DB     | 44.1  | 44.6  | 51.0  | 58.7  | 64.6  | 72.4  | 73.2  | 72.5  | 65.2  | 60.0  | 52.1  | 45.6  | (23) |
|                                                                        |             | MCWB   | 36.9  | 36.0  | 40.2  | 46.4  | 52.2  | 59.5  | 60.8  | 60.7  | 55.9  | 49.3  | 42.6  | 37.3  | (24) |
|                                                                        | 10%         | DB     | 41.4  | 41.6  | 47.5  | 55.0  | 60.9  | 68.7  | 69.9  | 69.0  | 61.9  | 56.8  | 48.9  | 42.9  | (25) |
|                                                                        |             | MCWB   | 35.2  | 34.4  | 38.5  | 44.3  | 50.6  | 58.1  | 59.5  | 59.4  | 54.1  | 48.5  | 41.5  | 36.1  | (26) |
|                                                                        | 0.4%        | WB     | 42.0  | 41.3  | 45.6  | 51.5  | 58.8  | 64.6  | 65.4  | 65.2  | 61.1  | 55.2  | 47.5  | 42.3  | (27) |
|                                                                        |             | MCDB   | 47.4  | 48.9  | 53.9  | 62.7  | 69.0  | 75.9  | 75.8  | 74.8  | 68.2  | 62.3  | 53.1  | 49.4  | (28) |
| (30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 39.7  | 38.9  | 43.1  | 49.1  | 55.8  | 62.5  | 63.1  | 62.9  | 58.7  | 52.9  | 45.5  | 40.4  | (29) |
|                                                                        |             | MCDB   | 44.7  | 44.8  | 51.1  | 59.4  | 64.6  | 73.0  | 72.8  | 72.5  | 66.1  | 59.2  | 51.9  | 46.5  | (30) |
|                                                                        | 5%          | WB     | 37.7  | 36.9  | 41.2  | 47.2  | 53.5  | 60.7  | 61.5  | 61.2  | 56.8  | 51.3  | 43.9  | 38.6  | (31) |
|                                                                        |             | MCDB   | 43.2  | 43.3  | 48.9  | 56.7  | 62.1  | 70.1  | 70.7  | 70.1  | 63.3  | 57.5  | 50.5  | 44.4  | (32) |
|                                                                        | 10%         | WB     | 35.5  | 34.9  | 39.3  | 45.4  | 51.5  | 58.9  | 60.0  | 59.7  | 55.0  | 49.5  | 42.0  | 36.8  | (33) |
|                                                                        |             | MCDB   | 40.6  | 41.1  | 45.8  | 53.8  | 59.4  | 67.3  | 68.4  | 67.7  | 60.5  | 55.2  | 48.4  | 42.0  | (34) |
| (36) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.8  | 12.9  | 14.1  | 15.6  | 15.4  | 15.7  | 15.8  | 15.1  | 14.3  | 14.0  | 11.9  | 11.8  | (35) |
|                                                                        |             | MCDBR  | 14.4  | 15.9  | 18.1  | 20.1  | 20.3  | 20.4  | 19.8  | 18.9  | 17.5  | 17.3  | 15.3  | 14.3  | (36) |
|                                                                        |             | MCWBR  | 8.8   | 9.3   | 10.4  | 11.4  | 11.4  | 11.3  | 10.8  | 10.7  | 10.3  | 9.4   | 8.3   | 8.2   | (37) |
|                                                                        | 5% WB       | MCDBR  | 13.4  | 14.8  | 16.4  | 18.8  | 18.6  | 19.1  | 18.6  | 17.8  | 16.2  | 15.5  | 13.9  | 13.7  | (38) |
|                                                                        |             | MCWBR  | 8.9   | 9.6   | 10.0  | 11.1  | 11.1  | 11.3  | 10.9  | 10.9  | 10.1  | 9.1   | 8.2   | 8.7   | (39) |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.248 | 0.274 | 0.318 | 0.358 | 0.376 | 0.379 | 0.377 | 0.365 | 0.342 | 0.317 | 0.280 | 0.248 | (40) |
|                                                                        | taud        |        | 2.342 | 2.281 | 2.235 | 2.234 | 2.261 | 2.297 | 2.295 | 2.341 | 2.398 | 2.448 | 2.452 | 2.383 | (41) |
|                                                                        | Ebn at Noon |        | 274   | 286   | 287   | 284   | 282   | 281   | 279   | 279   | 277   | 268   | 260   | 262   | (42) |
|                                                                        | Edh at Noon |        | 23    | 30    | 37    | 41    | 42    | 40    | 40    | 36    | 31    | 26    | 21    | 20    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 443   | 724   | 1080  | 1389  | 1576  | 1870  | 1846  | 1608  | 1264  | 821   | 465   | 361   | (44) |
|                                                                        | RadStd      |        | 50    | 101   | 154   | 198   | 137   | 176   | 158   | 145   | 110   | 86    | 63    | 56    | (45) |

## Historical Trends

|                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|--------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                          |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only             | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (112 neighbors) | N/A   | N/A     | N/A    | +1.31   | +0.74 | +0.56 | N/A         | N/A   | +103  | +29   |

Nomenclature: See separate page